

Problem 1:

12 & 7

$= 1100_2 \& 111_2$

$= 1100_2 \& 0111_2$

$= 0100_2$

$= 100_2$

$= 4$

problem 2:

$$15 \mid 10$$

$$\approx 1111_2 \mid 1010_2$$

$$\equiv 1111_2$$

$$\approx 15$$

Problem 3:

$$20^{13}$$

$$= 10100_2 \wedge 1101_2$$

$$= 10100 \wedge 01101$$

$$= 11001_2$$

$$= 25$$

Problem 4:

~ 10

$$= \sim (00001010_2)$$

$$= 11110101_2$$

$$= -128 + 64 + 32 + 16 + 0 + 4 + 0 + 1$$

$$= -128 + 117$$

$$= -11$$

Problem 5

$$8 \ll 2$$

$$= 1000_2 \ll 2$$

$$= 100000_2$$

$$= 32$$

Problem 6:

$$64 \gg 3$$

$$= 1000000_2 \gg 3$$

$$= 1000_2$$

$$= 8$$

Problem 7:

$$(5 \& 3) \mid 4$$

$$\Rightarrow (10_2 \& 011_2) \mid 100_2$$

$$\Rightarrow 00_2 \mid 100_2$$

$$\Rightarrow 10_2$$

$$\Rightarrow 5$$

Problem 8:

$$(10 \gg 1) \& 7$$

$$= (1010_2 \gg 1) \& 111_2$$

$$= 0101_2 \& 111_2$$

$$= 101_2 \& 111_2$$

$$= 101_2$$

$$= 5$$

Problem 9:

$$|^\wedge |^\wedge |$$

$$= (|^\wedge |)^\wedge |$$

$$= 0^\wedge |$$

$$= |$$

Problem 10:

$$\sim (-5_{10})$$

$$= \sim (2's \text{ comp}(5))$$

$$= \sim (2's \text{ comp}(0000010_2))$$

$$= \sim (1's \text{ comp}(0000010_2) + 1)$$

$$= \sim (1111010_2 + 1)$$

$$= \sim (1111011_2)$$

$$= 00000100_2$$

$$= 4$$

Problem 11:

$T \text{ or } F \text{ and } F$

$\equiv T \text{ or } (F \text{ and } F)$

$\equiv T \text{ or } F$

$\equiv T$

Problem 12:

$\text{not}(T \text{ and } F)$

$= \text{not}(F)$

$= T$

Problem 13:

$(5 > 3)$ and $(2 = 2)$

$\vdash T$ and $(2 = 2)$

$\vdash T$ and T

$\vdash T$

Problem 14:

0 or 5

= F or T

= T

[Because in python
0 = F and (number > 0 or
< 0) = T]

Problem 15:

10 and 20

= T and T

= T

(Because in python

0 = F and numbers < 0

0 = T)

Problem 16:

$\text{not} ("apple" == "orange")$

$= \text{not}(F)$

$= T$

Problem 17:

$$a = T, b = T$$

$$a \neq b$$

$$= T \neq T$$

$$= F$$

Problem 18:

128 circular << 1

= 10000000₂ circular << 1

= 00000001₂

= 1

Problem 19:

$$\text{bool}(C10)^1 \text{ bool}(C0)$$

$$= T^1 \text{ bool}(C0)$$

$$= T^1 F$$

$$= T$$

problem 20:

$$(7 \& 1) == 1 \quad \text{and} \quad (8 \gg 1) == 4$$

$$\Rightarrow (111_2 \& 001_2) == 1 \quad \text{and} \quad (1000_2 \gg 1) == 100_2$$

$$\Rightarrow 1 == 1 \quad \text{and} \quad 100_2 == 100_2$$

$$= T \quad \text{and} \quad T$$

$$= T$$

Problem 21:

$T \text{ or } F \text{ and not } T$

$\equiv T \text{ or } (F \text{ and } (\text{not } T))$

$\equiv T \text{ or } (F \text{ and } F)$

$\equiv T \text{ or } F$

$\equiv T$

Problem 22:

$$(23 > 42) \text{ and } (36 > 12) \text{ or } (39 \leq 39)$$

$$= F \text{ and } (36 > 12) \text{ or } (39 \leq 39)$$

$$= F \text{ and } T \text{ or } (39 \leq 39)$$

$$= F \text{ and } T \text{ or } T$$

$$= F \text{ or } T$$

$$= T$$

Problem 23 :

$$10 + 5 \gg 1$$

$$= (10 + 5) \gg 1$$

$$= 15 \gg 1$$

$$= 1111_2 \gg 1$$

$$= 11110_2$$

$$= 30$$

problem 24:

$A \text{ or } (\text{not } A \text{ and } B)$

$= (A \text{ or } \text{not } A) \text{ and } (A \text{ or } B)$

$= T \text{ and } (A \text{ or } B)$

$= A \text{ or } B$

Problem 25:

$(A \text{ and } B) \text{ or } (A \text{ and not } B)$

$= A \text{ and } (B \text{ or not } B)$

$= A \text{ and } T$

$= A$

Problem 26:

$A \text{ or } (F \text{ and } B)$

$\equiv A \text{ or } F$

$\equiv A$

Problem 27:

$\text{not } (\text{not } A \text{ and } B)$

$= (\text{not } (\text{not } A)) \text{ or } (\text{not } B)$

$= A \text{ or } \text{not } B$

Problem 2g:

$$5 \times 2^2 \& 15$$

$$= (5 \times (2^2)) \& 15$$

$$= (5 \times 4) \& 15$$

$$= 20 \& 15$$

$$= 10100_2 \& 01111_2$$

$$= 00100_2$$

$$= 4$$

Problem 29:

$\text{not}(15//3==5 \text{ and } 7\%2==0)$

$= \text{not}(5==5 \text{ and } 1==0)$

$= \text{not}(T \text{ and } F)$

$= \text{not}(F)$

$= T$

Problem 30:

$(A \text{ or } B) \text{ and } (\text{not } A \text{ or } B)$

$= (A \text{ and not } A) \text{ or } B$

$= F \text{ or } B$

$= B$

Problem 31:

$(A \text{ and } B \text{ and } C) \text{ or } (A \text{ and } B \text{ and not } C)$

$= A \text{ and } B \text{ and } (C \text{ or not } C)$

$= A \text{ and } B \text{ and } \top$

$= A \text{ and } B$

Problem 32:

$$\sim 5 \ \& \ 7$$

$$= (\sim 5) \ \& \ 7$$

$$= (\sim 00000101_2) \ \& \ 00000111_2$$

$$= 1111010_2 \ \& \ 00000111_2$$

$$= \begin{array}{r} \& \begin{array}{r} 1111010 \\ 0000111 \\ \hline 0000010 \end{array} \end{array}$$

$$= 0000010_2$$

$$= 10_2$$

$$= 2$$

Problem 33:

$$|_{\substack{< \\ 2}}^3 \mid |_{\substack{< \\ 2}}^1$$

$$= 100_2 \mid 001_2$$

$$= 101_2$$

$$= 10$$

Problem 34:

A and $(A \vee B)$

$\equiv A$

[Absorption property]

Problem 35:

$(T \text{ or } F) \text{ and } (\text{not } T \text{ or } T)$

$= (T \text{ or } F) \text{ and } (F \text{ or } T)$

$= (T \text{ or } F) \text{ and } T$

$= T \text{ and } T$

$= T$

Problem 36:

$A \text{ or } (A \text{ and } B) \text{ and } (B \text{ and } F)$

$= A \text{ or } (A \text{ and } B) \text{ and } F$

$= A \text{ or } F$

$= A$

Problem 37:

8 bit circular $\Rightarrow$ 1 by 1

= 8 bit circular $\Rightarrow$ 00000001₂ by 1

= 10000000₂

= 128

Problem 38:

$$10 > 5 > 2$$

$$\equiv (10 > 5) \text{ and } (5 > 2)$$

$$\equiv T \text{ and } (5 > 2)$$

$$\equiv T \text{ and } T$$

$$\equiv T$$

problem 39:

not $(5 < 2)$ or $10 / 2 == 5$

= not F or $10 / 2 == 5$

= T or $10 / 2 == 5$

= T or $5 == 5$

= T or T

= T

Problem 40:

$(A \text{ and } B) \text{ or } (\text{not } A \text{ and } \emptyset)$

(Let A and $B = c$)

$= (c \text{ or not } c)$

$= T$